

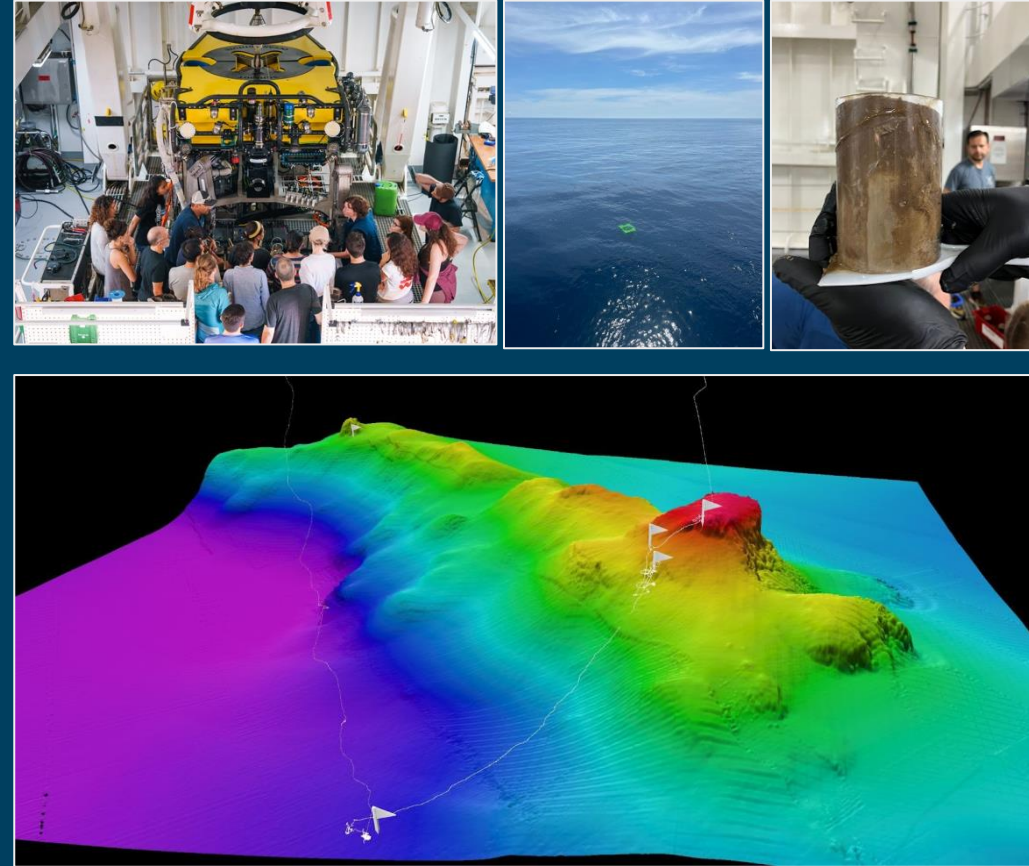
FKT Cd experiments dataset

Environmental Samples and Experimentation

Octopus Odyssey Expedition
Pacific Ocean, Costa Rica
June, 2023
Falkor(too)



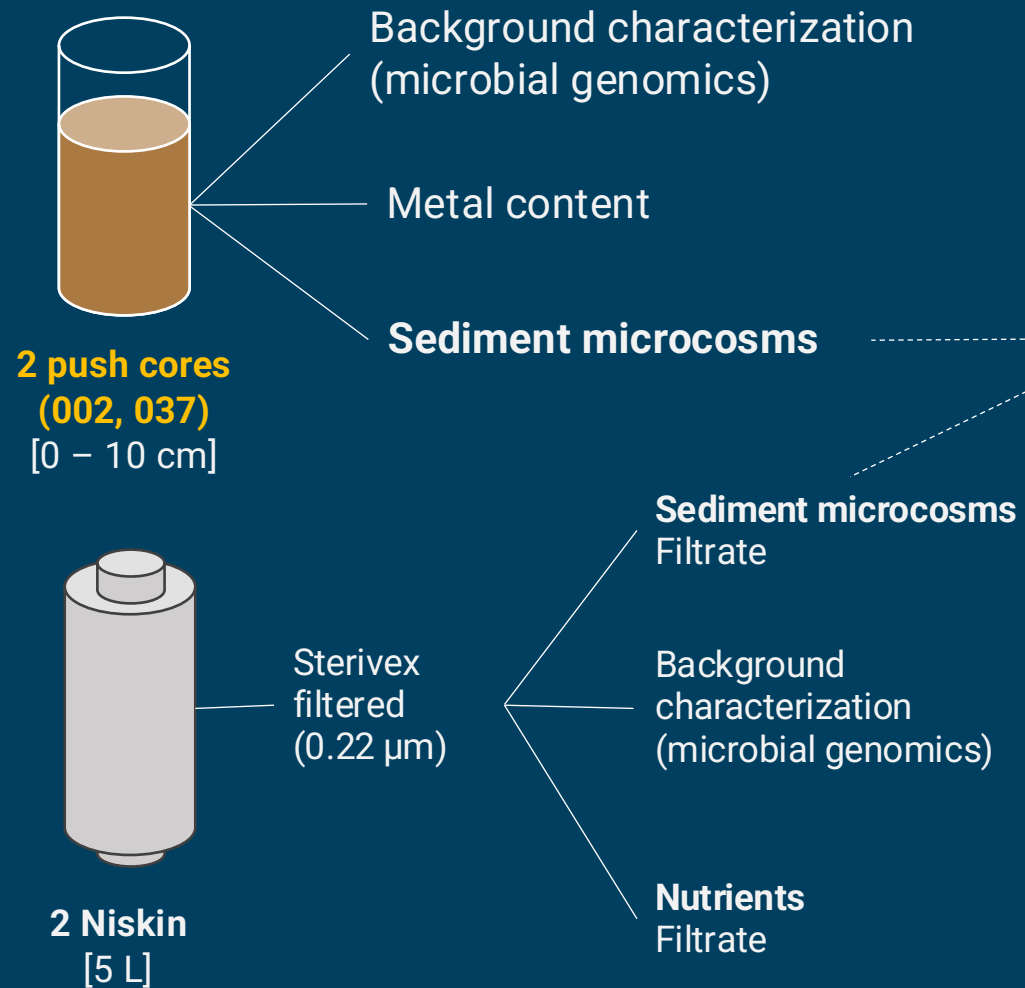
Photo credit: Alex Ingle (Schmidt Ocean Institute)



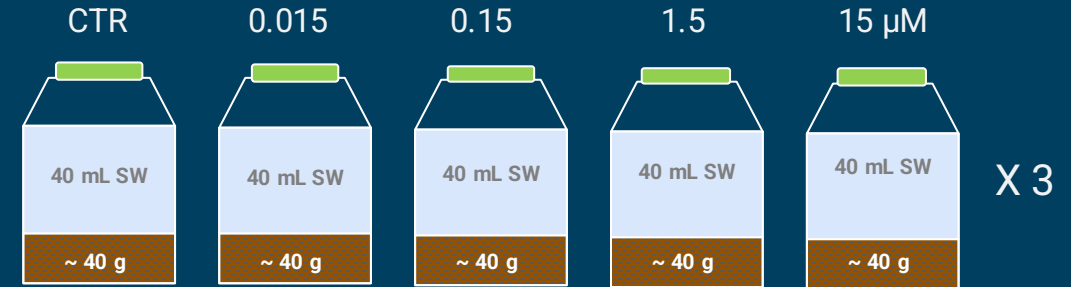
Sediments from five underwater seamounts (~ 3000m)

- Baseline community composition and diversity (taxonomic and functional)
- Responses to metal exposure (sediment microcosms with Cd)

From Dorado seamount (same location)



Sediment microcosms spiked with CdCl_2



After 96 hours incubation (4C) in the dark:

- Cd quantification (SW and sediments)
- Microbial genomics (DNA: **amplicon** and shotgun seq.)
- Relative expression (RNA: RT-qPCR and RNA seq.)

30 samples: 2 sediment cores x 5 Cd treatments x 3 biological replicates

16S sequencing analysis workflow (from short reads)

